PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
LONAE PRE-MOCK JOINT EXAMINATION

**FORM FOUR**

**GEOGRAPHY MARKING SCHEME** **MAY 2021**

**SECTION A**

1. **Multiple choice question. (10 items with 1marks @)**

| **Item** | **i** | **ii** | **iii** | **iv** | **v** | **vi** | **vii** | **viii** | **ix** | **x** |
|---|---|---|---|---|---|---|---|---|---|---|
| **Answer** | **B** | **A** | **B** | **D** | **A** | **B** | **D** | **A** | **A** | **C** |

2. **Matching item question (5 items with 1 marks @)**

| **Item** | **i** | **ii** | **iii** | **iv** | **v** |
|---|---|---|---|---|---|
| **Answer** | **B** | **D** | **C** | **F** | **A** |

**SECTION B (55 Marks)**

**Answer all questions in this section**

3. Study the statistical table bellow and answer the following questions

| Class interval | Frequency |
|---|---|
| 0-4 | 2 |
| 5-9 | 3 |
| 10-14 | 9 |
| 15-19 | 4 |
| 20-24 | 2 |

Calculate the following

i. Mean
ii. Mode
iii. Median
iv. Standard deviation

Solution

Table of calculation

| Class interval | Frequency (f) | Class mark (x) | fx | Deviation $\lvert x - \bar{x} \rvert$ | $d^2$ | $Fd^2$ |
|---|---|---|---|---|---|---|
| 0-4 | 2 | 2 | 4 | 10.25 | 105.06 | 210.12 |
| 5-9 | 3 | 7 | 21 | 5.25 | 27.56 | 82.68 |
| 10-14 | 9 | 12 | 108 | 0.25 | 0.062 | 0.55 |
| 15-19 | 4 | 17 | 68 | 4.75 | 22.56 | 90.24 |
| 20-24 | 2 | 22 | 44 | 9.75 | 95.06 | 190.12 |

$\sum f = 20$ $\sum fx = 245$ $\sum fd^2 = 573.71$

(i) mean (2 marks)

$\bar{x} = \frac{\sum fx}{\sum f}$

$\bar{x} = \frac{245}{20} = 12.25$

**Mean is 12.25**

(ii)mode (2.5 marks)

$L+\left(\frac{t1}{t1+t2}\right) i$

L=9.5
t1= 9-3=6
t2= 9-4=5
i= 5

$9.5+\left(\frac{6}{6+5}\right) 5 = 9.5+2.72 = 12.22$

**Mode is 12.22**

(iii)median (2.5 marks)

Data:

L=9.5
fb= 5
fw= 9
i= 5

median = $L+\left(\frac{\frac{N}{2}-Fb}{fw}\right) = 9.5+\left(\frac{\frac{20}{2}-5}{9}\right) = 9.5+\frac{25}{9} = 9.5+2.7 =$ **12.27**

**Median is 12.27**

(iv) Standard deviation (4 marks)

$= \sqrt{\frac{\sum fd^2}{\sum f}}$

$= \sqrt{\frac{573.71}{20}} = \sqrt{28.68} = 5.35$

**Sd = 5.35**

4. The form four students at Namanga Secondary School were told by their Geography teacher to measure the area of the school compound.
    i. Which survey method will be suitable for that measurement?
        - ✓ The suitable method is **CHAIN SURVEY**

ii. Four equipments that will be suitable for the method mentioned in (i) above: The student is required to explain each tool by giving the meaning, uses as well as diagram. These tools are:

- ❖ Chain, tape measure, ranging poles, arrows, pegs, surveyors band, cross staff, etc

iii. Two significances of the method mentioned in (i) above. The student is required to give brief explanation on any two importance of chain survey

- ❖ It is suitable for linear measurements
- ❖ Can be used in boundary and other government planning
- ❖ It is the cheapest method of survey since simple tools are used
- ❖ Uses few people in the field i.e leader, follower and booker
- ❖ Is used to fix the missing information in the existing plans and maps
- ❖ Can be combined with othe methods of survey like levelling, plane table and prismatic compass survey.

5. (i) Define the following terms as used in research:

a) **Research** is the scientific and systematic process of collecting information/data, analyzing and interpreting them in order to solve a certain problem.

b) **Case study**: this is the in-depth examinations of people or group of people or institution who should have similar characteristics. The case study method involves a careful and complete observation of a social unit, be that unit a person, a family, an institution, a cultural group or even the entire community.

c) **Research methodology**: is a way to systematically solve the research problem. It may be understood as a science of studying how research is done scientifically. In it we study the various steps that are generally adopted by a researcher in studying his research problem along with the logic behind them.

(ii) Explain any three research tools that can be used to investigate on the failure of geography exam at Longido Secondary School. These methods are as follows:

- Observation method
- Interview method
- Questionnaire method
- Focus group discussion

(iii) Briefly explain any five importance of research in daily life.

- ✓ Research helps to identify and solve various problems at different level. Research can be conducted to find the cause of a certain problem as well as suggesting the measures towards that problem. For example, on eruption of cholera, environmental pollution as well as on educational problems.

- ✓ Research is important since it provides a wide range of employment to people. To professionals in research methodology, research may mean a source of livelihood/income. This means some people do conduct research and sell them which enable them to get income.
- ✓ Research helps in policy and plan making. This can be to the individual institution or the government. For example on the introduction of industrial policy in Tanzania, the research must be done to find its possibilities, benefits as well as market where they can sell produced goods.
- ✓ To those students who are to write a master's or Ph.D. thesis, research may mean a careerism or a way to attain a high position in the social structure. Therefore research enable people to improve their professions.
- ✓ Research stimulated development. Increased amounts of research make progress possible. Research inculcates scientific and inductive thinking and it promotes the development of logical habits of thinking and organisation. This means that, all kinds of development is the product of research though inquiry/investigation and logical thinking.
- ✓ Research helps to learn/study more about things, people and events. For example to study peoples behavior through qualitative research, to study cause and effects of events like occurrence of road accidents. Therefore it helps in making decisions.
- ✓ Research help to get new body of knowledge about certain phenomena. Through research is where we can obtain new knowledge concerning a certain thing through logical reasoning and investigation. For example on water crisis, mass failure in secondary schools, etc.
- ✓ To philosophers and thinkers, research may mean the outlet for new ideas and insights. This means through research is where philosophers can develop new concepts or re-interpret the existing ones.
- ✓ Research is also used to establish new theories as well as for testing the existing theories.

6. Study carefully the photograph bellow and then answer the questions that follow



(i) The type photograph above is **GROUND/HORIZONTAL PHOTOGRAPH** because it shows the horizon, small area is covered, have no constant scale because objects in fore ground appear bigger than middle and background, etc

(ii) The common name of the animals seen on the photograph is **LION.**

(iii) Two human activities that are likely to affect the animals mentioned in (ii) above are ***illegal hunting, *agricultural activities, *deforestation**, etc

**(iv)** The main economic activity likely to take place in the photographed area is **TOURISM ACTIVITY**

(v) Two economic benefits of the activity mentioned in (iv) above are ***employment**, ***source of national income**, ***social services provision**, etc

(vi) Two areas where this photograph might have been taken in Tanzania are *Tarangire N.P, *Ngorongoro conservation area, *Manyara N.P, *Mikumi N.P, Etc

7. Study careful the map extract of SIKONGE (sheet 137/2) and answer the questions that follows
   (i) To give two supportive contents that have been used on the given map by giving example. (2marks)
      a. Tittle eg SIKONGE
      b. Scale eg linear scale and RF scale shown is 1:50,000
      c. Key found in southern part of the mapped area.
      d. Date of compilation which is 1978, there is margin, etc
   (ii) The area covered by the whole map. Give your answer in $Km^2$ (use grid square method)
   Data
   Full squares=143
   Half squares=52
   Area =full squares+ $\frac{half\ squares}{2}$ = 143+$\frac{52}{2}$ =143+26 = 169 squares
   From the scale 1:50,000
   1cm on the map represent 0.5km on the ground, therefore since 1 square =2cm each sides
   1square = $1km^2$
   169 square = ? = $\frac{169sq \times 1km2}{1sq}$ = $169km^2$
   **<u>Therefore the area of the whole map is $169km^2$</u>**
   (iii) With evidences from the given map, give any two methods used to show relief of the mapped area.
      a. Triangle or trigonometric ratio found at grid reference 795849 showing height of a hill which is 1229m
      b. Contour method shown almost in all parts of the map
      c. Spot height shown in many parts of the map for example in SW of the map .1126
   (iv) Comment on settlement pattern by referring on the given map.
      a. Nucleated settlement pattern in SW part of the map
      b. Scattered in many parts of the map like in central part at Kisanga
      c. Linear settlement pattern in north eastern part of the map
   (v) By giving evidence from the given map, comment on any two (2) economic activities carried on the mapped area.
      a. **Trade activities** due to presence of town area at South western part of the ma and presence of road passing though Iyombakuzaa area to Sikonge
      b. **Agriculture activities** due to presence of scattered cultivation an many parts of the map for example in central part of the map.
      c. **Lumbering or forestry activities** due to presence of woodlands in almost all parts of the mapped area for example in north western part.

## SECTION C (30 Marks)

**Answer any two questions from this section**

8. The late President John Pombe Mgufuli had a dream to make Tanzania an industrialized country. What do you think are the challenges which President Samia Suluhu Hassan should work upon for the dream to become true? (6 points)
   - **Introduction: It should include the meaning of manufacturing insustry, its types, some industries found in Tanzania, etc. (1.5 marks)**
   - **Main body: the student is supposed to give clear six point, good explanations and supportive examples (2 marks @)**

i. Shortage of labour.
ii. Fund or capital problem
iii. Shortage of power and energy
iv. Poor transport and communication networks.
v. Shortage of market for manufactured goods

vi. Price fluctuation which discourages industrial producers.
vii. Lack of strong support from the government
viii. Shortage or scarcity of water
ix. Poor availability of raw materials
x. Competition from other countries producing the same products or goods. For example China competition with South Korea on electronic equipment.

- **Conclusion: any relevant conclusion relating to the asked question (1.5 marks)**

9. Population distribution and density is uneven through the world. What are the reasons behind this dispersion in Africa? (6 point)
   - **Introduction: It should include the meaning of population, population distribution and population density (1.5 marks)**
   - **Main body: the student is supposed to give clear six point, good explanations and supportive examples (2 marks @)**

i. Relief (Topography) Where the slope is steep there is low or no population due to poor soil and nature of the land, but on gentle slope or flat surface there is high population since soil is good and mechanization can take place easily. Also low land tends to flood usually since people avoid settling in those areas.
ii. Climate. Consider temperature and rainfall
iii. Vegetation Dense vegetation hinders penetration and development. In less dense vegetation people are attracted since they grow crops after clearing for cultivation.
iv. Soil or edaphic factor. Thin, infertile and badly leached soil discourages settlement since they can't support agriculture. Good soil attracts population since they can practice agriculture.
v. Mineral and energy resources Area with mineral and energy resources attract population example Mwadui, Mererani and Buzwagi in Tanzania.
vi. Natural Hazards Areas with natural hazards like floods in low land, earthquakes, and tornadoes are avoided by people and low or no population on those areas.
vii. Diseases and Pests People like settling in areas which are free of diseases and pests.
viii. Social – cultural aspects. Traditional beliefs like superstitions or witchcraft can make people avoid living in certain areas due to fear of risking their life. Presence of relatives
ix. Area where social services are ready available like in towns, will have many people unlike the rural areas where social services are poorly available. Eg schools, hospitals, water services, power services as well as transport and communication.
x. Political factors Area with political stability and peace attract population but where there is political instability does not attract population. Economic factors.
xi. Employment opportunities attract many people to settle on that area than areas where there is no such opportunities.
xii. Trading activities also influence population of a place since the area where people can engage in trade will have many people compared to other areas.

- **Conclusion: any relevant conclusion relating to the asked question (1.5 marks)**

10. Climate is the only factor for soil formation. Discuss this statement.
    - **Introduction: It should include the meaning of soil, types of soil as well as components of soil (1.5 marks)**

Soil is the uppermost surface layer of loose or unconsolidated material which overlies the crystal rocks and on which plants grow OR Soil is the natural body of mineral, organic and nutrient constituents which result from the interaction of the country rock with the environmental factors of climate, topography, plant and animal life.

- **Main body: the student is supposed to give clear five point, good explanations and supportive examples (2.5 marks @)**

## *Factors for soil formation*

The type of soil and rate of soil formation depend on a number of interacting factors in a particular environment, hence soil is the product of its own environment. Therefore climate is not a only factor for soil formation but involves a number of factors including climate itself.
These factors are ***parent materials, climate, living organisms, topography and time***:

i. **Parent Material (Rock):**This determines the type of soil formed, soil colour, soil depth, the rate of soil formation, soil structure, soil texture, porosity and mineralogical composition or its fertility. It also influence soil maturity, such that if the rock is hard, It takes a long time for soil to mature while the rate of maturity is fast where the parent rock is soft.

ii. **Living Organisms (Biotic):**These include the influence of plants, animals as well as human beings. vegetation contribute to soil fertility by adding humus in the soil after dying and decomposing. Burrowing animals facilitate weathering process by loosening the soil particles. Lastly man's activities like cultivation, break up the rocks into smaller fragments.

iii. **Climate**: The important variables under climate include temperature, precipitation and wind. Temperature affects the rate of decomposition of organic matter, it contribute to soil profile development through weathering as well as the rate soil development. Precipitation also affects soil profile development. Rainfall adds moisture, which facilitate both chemical and mechanical weathering and hence soil profile development.

iv. **Relief (topography):** topography influences climate and vegetation. It controls the rate and nature of weathering, removal and deposition of the soil parent materials. Soil erosion is rapid on steep slopes and less on gentle slopes. Therefore on steep slopes soil profiles are shallow while on gentle slopes they are expected to be deeper. In terms of aspects, the side that receive more sunshine tends to have well developed soil than the side which receives low amount of sunshine since isolation accelerate plants growth and the weathering process.

v. **Time**: The time factor may be associated with the age of the country rock especially those on which soil have been directly formed. It may also be associated with the duration of the operation of soil formation process, that is whether the soil has sufficient time to form mature profiles and associated characteristics or not. When soil formation has taken a long and enough time, soil tends to be more mature, they are usually deep and well developed.

- **Conclusion: any relevant conclusion relating to the asked question (1 marks)**